

Multiple Schedulers in Kubernetes

Since you're studying **KCNA**, let's understand this in a beginner-friendly way.

1. First: What is a Scheduler?

The **Kubernetes Scheduler** decides **which Node should run a Pod**.

For example:

Pod

|



Scheduler

|

├— Node 1 ❌ Busy

├— Node 2 ✅ Suitable

└— Node 3 ❌ Not suitable

|



Node 2

So remember:

📌 **Scheduler = Decides where a Pod should run.**

2. What are Multiple Schedulers?

Normally, Kubernetes has **one default scheduler**:

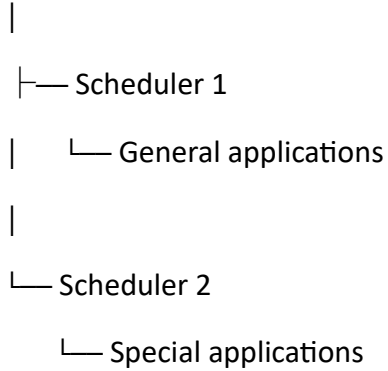
kube-scheduler

But Kubernetes can also have **more than one scheduler**.

Multiple schedulers means running multiple scheduler processes in the same Kubernetes cluster, where different Pods can be assigned to different schedulers.

For example:

Kubernetes Cluster



3. Why Use Multiple Schedulers?

Sometimes the default scheduler's rules are not enough for a particular application.

You might want a special scheduler for a particular type of workload.

For example:

Scheduler 1



Normal applications

Scheduler 2



Special applications

This allows you to have **different scheduling policies** for different workloads.

4. Real-Life Example 🏠

Imagine a school has two people assigning students to classrooms.

Scheduler 1 👤

Handles normal students:

Students → Normal classrooms

Scheduler 2

Handles special students:

Special students → Special classrooms

Both are doing the same general job:

Deciding where students should go.

But they use different rules.

That's similar to **multiple schedulers**.

5. How Does a Pod Choose a Scheduler?

A Pod can specify which scheduler should handle it using:

spec:

 schedulerName: my-scheduler

For example:

apiVersion: v1

kind: Pod

metadata:

 name: my-pod

spec:

 schedulerName: my-scheduler

 containers:

 - name: nginx

 image: nginx

Here:

schedulerName: my-scheduler

means:

"I want my-scheduler to schedule me."

6. Default Scheduler

If you don't specify a scheduler:

spec:

containers:

Kubernetes normally uses:

default-scheduler

So:

No schedulerName



Default Scheduler

But:

schedulerName: my-scheduler

means:

Pod



my-scheduler

7. Example with Two Schedulers

Imagine we have:

Scheduler 1

default-scheduler

Scheduler 2

gpu-scheduler

Normal Pod

spec:

 schedulerName: default-scheduler

Goes to:

default-scheduler

GPU Pod

spec:

 schedulerName: gpu-scheduler

Goes to:

gpu-scheduler

8. Important Point ★

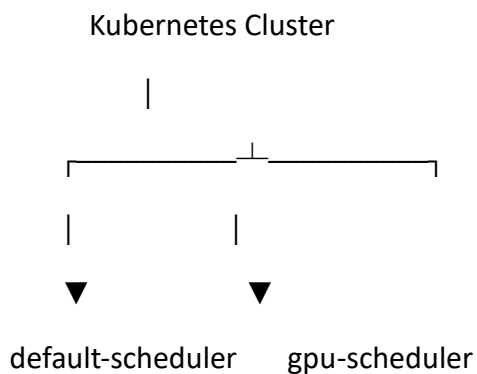
Multiple schedulers **do not mean that all schedulers schedule every Pod.**

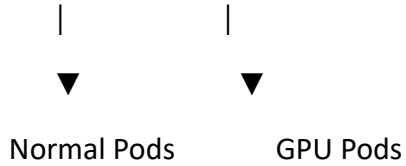
Instead, a Pod is associated with a particular scheduler through:

 schedulerName:

The scheduler whose name matches the Pod's schedulerName handles that Pod.

9. Visual Understanding





10. Why Is This Useful?

Multiple schedulers can be useful when you need:

- Different scheduling policies
- Special workloads
- Custom scheduling logic
- Different rules for different applications

For example:

Normal applications



Default Scheduler

GPU/AI applications



Custom Scheduler

11. Multiple Schedulers vs Node Affinity

Don't confuse them.

Node Affinity

You tell Kubernetes:

"I prefer or require a Node with these labels."

Pod



Scheduler



Node Affinity rules



Suitable Node

Multiple Schedulers

You tell Kubernetes:

"Use this particular scheduler to decide where I should run."

Pod



schedulerName



Specific Scheduler



Suitable Node

KCNA Quick Revision ★

Scheduler



Decides **where Pods should run**.

Multiple Schedulers



Allows **more than one scheduler** to operate in a Kubernetes cluster.

schedulerName



Specifies **which scheduler should schedule a Pod**.

Example:

spec:

```
  schedulerName: my-scheduler
```

Default

If no scheduler is specified:

default-scheduler

is normally used.

Easy Memory Trick

ONE Scheduler



Most Pods



default-scheduler

MULTIPLE Schedulers



Different Pods



Different scheduling policies

One-line KCNA definition:

Multiple schedulers allow a Kubernetes cluster to use different scheduling components so that different Pods can be scheduled according to different policies.